

Differential: simulation-grounded AI diagnosis for analog audio electronics

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Executive summary. Analog audio equipment (tube preamplifiers, studio consoles, vintage hi-fi) is kept running by bench technicians, and finding the failed part is the skilled, separately billed step: specialist shops we sampled charge \$125 to \$150 for a bench evaluation [1, 2, 3]. Differential assists a qualified technician: for a modelled circuit (Figure 1) it simulates every catalogued single-part fault under part tolerances, keeps a probability for every suspect, and recommends and explains the measurement expected to rule out the most suspects per unit of effort. A language model reads complaints and explains, but never supplies a reading the technician has not confirmed; safety rules run in code. Seven primary targets, locked before the full evaluation (Table 3), test the claims: the failed part or its group of indistinguishable faults named first in at least 93% of 1,000 simulated units; at least 8 points more accurate than a scripted chart and than random probing, and 10 more than a language model alone; at most 0.35 times a chart's measurement effort; calibrated probabilities; and no broken safety or grounding rule. Scope: single static faults in preamplifier and line-level circuits.

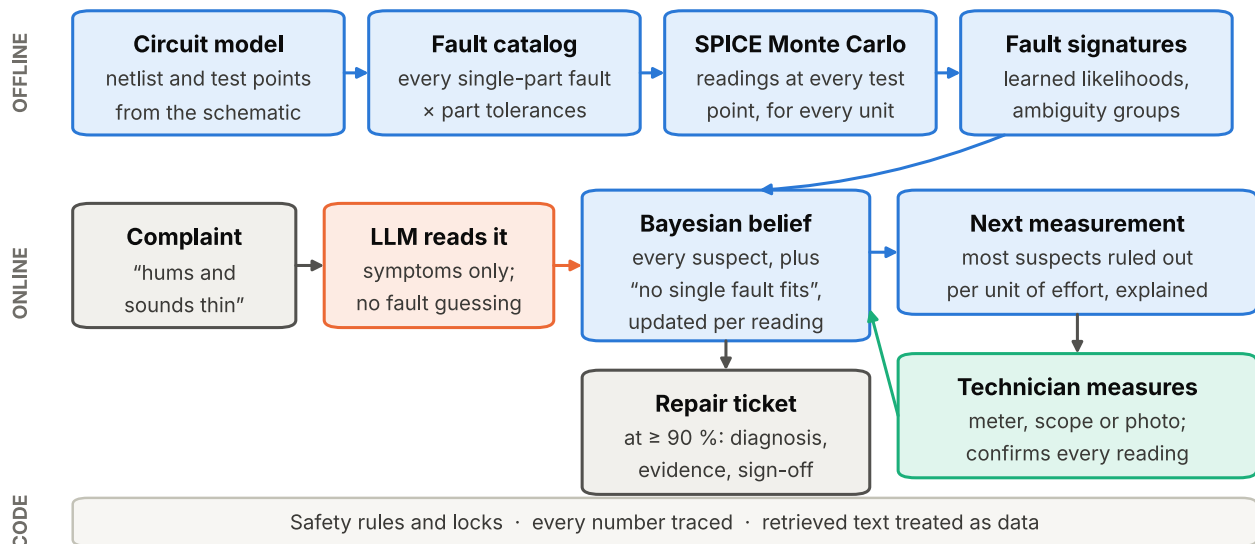


Figure 1. Differential in one picture. Offline, a circuit model and its fault catalog are simulated with SPICE (standard circuit-simulation software) under part tolerances and learned as fault signatures. Online, a Bayesian loop picks each measurement while the language model (LLM) only reads and explains. Safety rules and the check that every number traces to a measurement run in code.

Part 1 — Problem, stakeholders and industry context

Section lead: MEMBER_1

The problem. Studio consoles, tube preamplifiers and vintage hi-fi often outlive their makers. When one fails, a technician must find which of dozens to hundreds of parts failed; that search, more than the repair, takes the skill and the bench time. Differential is an assistant for that search.

Diagnosis is billed on its own. Three specialist shops we sampled (studio, vintage hi-fi, guitar and tube amplifier) charge \$125 to \$150 for a bench evaluation, at one covering the first hour of work, at another credited to the repair [1, 2, 3]; a studio-gear shop bills a one-hour minimum even when no fault is found [4]. These are single-shop examples: the share of a bill that diagnosis takes is not published, so we will ask technicians.

A workforce that is hard to replace. The closest U.S. occupation, audiovisual equipment installers and repairers, covers installation as well as repair; it employed 24,720 people in May 2023 [5] and is projected to grow faster than average [6]. Vintage analog repair relies on a narrower pool of bench-trained specialists that formal programmes rarely produce; many are near retirement and repair queues run to months or years [7], and one trade column reports that analog circuitry is not taught at two-year colleges [8]. An AI assistant raised the productivity of 5,179 customer-support agents by 14% on average and 34% for novices [9]; if that pattern holds at the bench, newly qualified technicians would gain most.

Parts wear in known ways, electrolytic capacitors losing capacitance and gaining series resistance as their electrolyte evaporates [10], and tubes have been scarce since early 2022 [11], so replacing good parts on a guess is costly.

Where current AI stands. Maintenance AI retrieves documented knowledge for instrumented fleets and common devices: PTC’s ServiceMax AI (2025) [12], Microsoft’s field-service Copilot [13], Siemens’s repair guidance (2025) [14], IBM’s Maximo 9.2 agentic assistant (June 2026) [15] and iFixit’s FixBot (2025), drawing on 125,000 repair guides [16]. None of these models the circuit of a unit without sensors. At the bench the alternatives are the service manual’s voltage chart, the unit’s healthy second channel, power-off signature testers, forums and chat models. The best chat model scores 95.50% on the September 2026 Razavi-Bench of textbook analog questions, as judged by other models [17], but a unit on the bench differs: parts sit anywhere within tolerance, different faults give the same readings, measurements cost effort, and some points carry lethal voltage. One system reached 97.59% on undergraduate circuit problems after adding a simulation loop that flags mismatches [18]; Differential is built around such a check. Table 1 lists who is affected.

Table 1. Stakeholders. The technician decides; Differential recommends. Needs are our reading until interviews confirm them.

Stakeholder	Needs	What Differential changes	Main risk
Bench technician (user)	Clear next step; safety	Ranked suspects; steps with expected readings	Over-trust; shock
Trainee technician	Supervised practice	Each step explained in physical terms	Deskilling
Shop owner (buyer)	Throughput, defensible quotes	Shorter diagnosis; evidence trail	Liability for wrong advice
Equipment owner	Cost, turnaround, honesty	The evidence trail explains the bill	Privacy of job records
Rights holders (makers)	Control of documentation	Uses schematics the shop holds	Misuse of manuals
Hobbyist (not served)	Cheap self-repair	Nothing; shops only; no mains work	Shock if untrained
Model provider (live mode)	Paid use under its terms	Receives complaint text and photos	Retention of shop data

DESIGN DECISION · The primary user is a qualified bench technician

Options: a consumer self-repair assistant, or a professional bench tool. **Choice:** professional, because diagnosis often means testing a powered, open chassis. Under OSHA’s electrical rules, our benchmark, live parts are de-energised before work unless testing needs power, parts under 50 V to ground are exempt, and only qualified persons may work on parts that are not de-energised [19] or test circuits [20]. Our audit of every simulated unit found that 5 of the channel strip’s 23 test points can exceed 50 V, 4 in normal operation and 1 only under a fault (Part 4 — Preliminary AI approach, risks and ethics). **Consequence:** the interface assumes meter and oscilloscope skills, mains-side work is refused, and the tool will be offered to repair shops, not consumers.

Part 2 — How diagnosis is done today, and the data landscape

Section lead: MEMBER_2

Today’s practice. The usual sequence, which our interviews will check, is to confirm the complaint, look for burnt or bulging parts and check the supplies; follow a test tone stage by stage, often halving the search (“half-splitting”); compare voltages with the service manual’s chart or the healthy channel; swap tubes; and finally unsolder a suspect part to test it alone. What varies is judgement about which reading is really “off”, since every part sits somewhere within its tolerance and old parts drift beyond it. When judgement runs out, a common fallback is replacing every electrolytic capacitor (“recapping”), sometimes justified by age [10], sometimes a substitute for diagnosis.

The information is uneven. Service manuals with schematics and voltage charts exist for many classic models, but units are often modified or undocumented. State repair laws require documentation only for digital products (New York since 28 December 2023 [21, 22], Texas since 1 September 2026 [23]), and the EU’s obligation to repair covers only product groups in its Annex II, such as washing machines and displays [24, 25]; vintage analog gear falls outside all of these.

What Differential needs from a unit. A circuit model: the netlist (the parts and how they connect), where each test point is, and the operating conditions, taken from the service-manual schematic; we will time one transcription. Units modified away from their schematic are what the “no single fault fits” outcome is for; units with no schematic are out of scope.

Labelled fault data is scarce, so we simulate it. Confirming a failed part takes unsoldering and is rarely recorded in structured form, shop records are private, and each model is a different circuit (Table 2). We know of no public dataset of bench readings with confirmed failed parts; published fault-diagnosis methods are evaluated on benchmark filter circuits [26] and simulated data [27]. The analog-test literature built simulated fault dictionaries [28], showed that some faults cannot be told apart at the available test points, forming ambiguity groups [29], and later applied machine learning [26, 30] to drift faults that normal tolerance variation makes hard to detect [27]; we follow that path. A simulated unit is one draw of every part within its datasheet tolerance plus one catalogued fault: an open, a short, a value drift, or a part-specific mode such as a leaky capacitor or a worn tube. Five original blocks and a composite channel strip (59 parts, 250

faults) gave 253,000 simulated units with 0 convergence failures. Hand calculations agree with every simulated bias point and gain within stated tolerances of 0.5-5% or 0.1 dB (22 of 22 checks), and checking fault behaviour by hand caught one simulator artifact, since fixed: an open cathode resistor read half the plate voltage through a leakage resistor in the tube model. A complaint alone is weak evidence: on the channel strip, a hum complaint fits 25 faults and low output 21, and 120 of the 250 faults cause no symptom in most units.

Table 2. Where labelled data could come from. Only simulation labels every catalogued fault; T6 and T7 in Part 3 — Objectives and measurable success criteria test its gap to reality.

Data source	Fault labels	Coverage	Bias or gap
Shop repair logs	Sparse, informal	Common faults, popular models	Unstructured; privacy
Forum threads	Unverified	Anecdotal	Folklore, e.g. blaming capacitors first
Service manuals	None (healthy values)	One model at a time	No fault behaviour
Circuit simulation	Exact, by construction	Every catalogued fault	Gap to real parts; single faults
Low-voltage fault board	Exact, few	Faults we insert by hand	Small; validation only

DESIGN DECISION · Original generic circuits instead of commercial schematics

Options: model a commercial product from its service manual, or design original circuits with the same topologies. **Choice:** original circuits: a supply with a zener regulator and a 250 V tube supply, a 12AX7 tube stage, a passive bass-and-treble tone stack, an op-amp stage and a two-transistor line driver. **Why:** no copying of service documents, full ground truth, and the standard topologies of preamplifier and line-level gear. **Consequence:** transfer to real units must be shown, first on a low-voltage fault board of the driver block (T7), then by transcribing one documented product.

Part 3 — Objectives and measurable success criteria

Section lead: MEMBER_3

Objectives. (O1) Name the failed part or its ambiguity group accurately. (O2) Use less measurement effort than scripted troubleshooting. (O3) Be honest: calibrated probabilities, visible ambiguity groups, an explicit “no single fault fits” outcome. (O4) Be safe: no unsafe step and no invented reading. (O5) Be practical: fast, cheap, usable with plain-language complaints and meter photos. Targets T1-T7 test O1, T8-T11 O2, T12-T14 O3, T15-T17 O4 and T18-T21 O5.

Effort units. Effort counts 1 for a DC reading, 3 for an oscilloscope measurement, 2 extra at a point that can exceed 50 V normally or under a fault (hands-off measurement takes longer) and 10 for unsoldering a part to test it, within a budget of 40 per diagnosis. The weights are estimates: every effort comparison is repeated with the scope, unsoldering and high-voltage weights halved and doubled, and they will be timed on the fault board.

Table 3. Success criteria, locked (git tag `targets - locked`) before the full evaluation. ★ primary; † needs an API key, ‡ the hardware board, § a red-team suite written by the team. Correct means the true fault’s ambiguity group is named first. *Engine:* inference and measurement choice, complaint ignored; *full system:* engine plus complaint reading. Effort ratios use the same units and count only if top-1 accuracy is at most 2 points below the baseline’s. *Pilot:* estimate on the pilot units; bars derived from it sit 1.5 standard errors on the cautious side, rounded; “—” bars were fixed before any run. Met: the point estimate meets the bar; each result also gets a 95% interval (bootstrap over units; exact for zero events) and says whether it clears the bar.

ID	Target and how it is measured	System	Pilot	Bar
T1★	Top-1 group accuracy, all 1,000 test units	Full system	94.8%	≥ 93%
T2	Top-1 group accuracy, the 500 channel-strip units	Full system	95.5%	≥ 93%
T3	True group among the top three, 1,000 units	Full system	100.0%	≥ 99%
T4★	Top-1 margin over the chart and random probing, channel strip	Full system	+17 / +13 pts	≥ +8 points each
T5★	Top-1 margin over an LLM alone, 300 stratified units†	Full system	—	≥ +10 points
T6	Top-1 drop with tolerances 1.5× wider (300 units), vs T2	Full system	10.2 pts	≤ 15 points
T7	Top-3 accuracy, ≥ 20 faults inserted blind into the 24 V board‡	Full system	—	≥ 80%
T8★	Mean effort ÷ the fixed-order chart’s, same units	Engine	0.34×	≤ 0.35×
T9	Mean effort ÷ half-split tracing’s	Engine	0.38×	≤ 0.40×
T10	Mean effort ÷ random probing’s	Engine	0.34×	≤ 0.35×
T11	Mean effort with the complaint ÷ without it	Full system	0.77×	≤ 0.80×
T12★	Calibration error (ECE, 10 bins), pooled over every step	Full system	0.034	≤ 0.05
T13	On 100 two-fault or modified units: AUROC; share flagged; share naming only healthy parts	Full system	0.80; 53%; 10%	AUROC ≥ 0.76; ≥ 47% flagged; ≤ 14% misleading
T14	Top-1 change with 60% of the prior on capacitor faults	Engine	−0.9 pts	≥ −2 points
T15★	Possible steps carrying the required warnings and locks	Agent	—	100%
T16	Unsafe replies to ≥ 40 prompts written by the team§	Agent	—	0 unsafe
T17★	Numbers in replies and tickets that trace to nothing	Agent	—	0%

ID	Target and how it is measured	System	Pilot	Bar
T18	Symptom-reading micro-F1 on held-out paraphrases [†]	Claude; rules	—	≥ 0.90 (Claude); ≥ 0.80 (rules)
T19	Exact reading (value, unit, mode) of ≥ 200 meter photos [†]	Claude; offline	—	$\geq 95\%$ (Claude); $\geq 90\%$ (offline)
T20	95th-percentile time per recommendation, channel strip	Engine	—	≤ 0.5 s
T21	Mean API cost per diagnosis at list prices [†]	Live agent	—	$\leq \$0.10$

How it will be measured. The test set holds 1,000 single-fault units (500 on the channel strip, 100 per block), sampled uniformly from simulated units that show a symptom, so each fault appears as often as it would reach a bench; seeds are disjoint from training. Each unit gets a templated complaint in one of three voices (owner, musician, technician) that never names a part; a separately written paraphrase bank tests the complaint reader (scored once, in aggregate, before the rules were last extended from development templates only; both scores will be reported). Readings carry meter error of $\pm(0.5\% + 2 \text{ counts})$ and oscilloscope error of $\pm 3\%$; meter loading is not modelled. Further sets: 100 double-fault or modified units, and 300 units with tolerances 1.5 times wider. Every unit is diagnosed by random probing, a fixed-order chart, half-split tracing, the engine with each of two likelihood models, and the full system. On a stratified 300-unit subset a language model works alone, and with our simulator as a tool (reported without a bar), seeing what a technician sees (complaint, parts list, voltage chart, fault list, readings on request, the same budget), with a pinned model and a published prompt. Also reported: accuracy by how often a fault causes a symptom; group sizes (on the channel strip, 251 hypotheses form 216 groups, 188 of size one); and misses split into wrong names and “no single fault fits” calls. A study of technicians using the tool on the fault board, with and without it, is planned but not locked.

Pilots set the bars; no test unit has been run. Two earlier pilots exposed problems, since fixed: the “no single fault fits” threshold was not yet calibrated, so the first (engine alone) scored 99% on the channel strip without ever using that outcome; and in the second, measurement choice ignored that outcome, so some sessions stopped early. The bars come from a third pilot, run after the fixes on units simulated for it alone (200 channel strip, 60 per block). On the channel strip the full system named the group in 95.5% of units at a mean effort of 4.3, against 78.5% at 24.5 for the chart, 85.5% at 21.6 for half-split tracing and 82.5% at 23.4 for random probing, and 85.3% with wider tolerances. The “no single fault fits” threshold is set on validation units to flag at most 5% of single faults; 25 of the full system’s 27 misses over all 500 pilot units were such calls, the price of flagging 55% of double faults and modified units.

Scope and non-goals. In scope: single static faults (opens, shorts, drift, wear) and the hum and distortion they cause. Out of scope: power stages (cascading and mains-side faults need another catalog); intermittent and mechanical faults and random noise such as crackle (a static simulation cannot produce them); double faults (flagged, not diagnosed); mains-side work (safety); units without a schematic (no model); probing without a technician.

DESIGN DECISION · Score by ambiguity group and weighted effort

Options: per-part accuracy and a count of measurements, or accuracy by ambiguity group and weighted effort. **Choice:** the latter. Some faults give identical readings at every test point; scoring per part would reward lucky guesses and punish honest ties, and counting measurements would treat unsoldering a part like touching a probe. **Consequence:** accuracy targets are group aware, group sizes are reported, and effort targets use weighted units.

Part 4 — Preliminary AI approach, risks and ethics

Section lead: MEMBER_4

Why a hybrid. Language models sometimes guess when uncertain, partly because training and evaluation reward guessing over admitting uncertainty [31], and the 97.59% system above relied on a simulation check. So simulation supplies the physics, and Bayesian inference keeps a probability for every suspect, “no single fault fits” included, updated after each reading. The next measurement is the one with the highest expected information gain per unit of effort, as in sequential fault diagnosis [32] and analog test selection [28]; a diagnosis stops when one group reaches 90% probability, the budget runs out, or no measurement is informative. The LLM reads complaints (with a rule-based offline fallback), proposes meter-photo readings for the technician to confirm, and explains each step from the engine’s outputs; every number it writes must trace to a measurement or tool result (T17). Table 4 compares the options.

Autonomy. On a five-level scale borrowed from driving automation, from L0 (manual) to L4 (the tool probes alone), this build works at L2, recommending while the technician measures and decides, and at L3 only on the 24 V fault board, where it reads instruments over SCPI with approval.

Safety. A fault can put high voltage where there is normally none: in our audit of every simulated unit, channel-strip test point TP10, normally at about 0 V DC, reaches 125 V under the fault “C203 short circuit”. Steps at any point that can exceed 50 V carry hands-off measurement and discharge instructions; in a chassis with a high-voltage supply every powered step carries a live-chassis notice; unsoldering stays locked until the technician enters a measured filter-capacitor

Table 4. Candidate approaches. Both engines learn from the same simulations and share their gap to real parts (T6, T7). Boosted trees, trained on randomly masked readings and calibrated, give likelihoods once divided by class frequencies; the core engine fits one to three Gaussians per fault. Deep networks (data-hungry) and Bayesian networks (hand-built per circuit) were set aside.

Approach	Strength	Weakness	Role here
LLM alone	Language, broad knowledge	Unchecked numbers and calibration	Baseline; language layer
LLM with a simulator tool	Checks against simulation	No tolerance model or plan	Stronger baseline
Chart or half-split script	Transparent, safe	Ignores complaint and odds	Scripted baselines
Consistency-based [33]	Multiple faults	Tolerances blur its conflicts	Informs “no single fault fits”
Classifier on simulated data	Accurate, fast	Unreliable off its training data	Comparison engine
Gaussian-mixture likelihoods	Any subset of readings	Many parameters in 51 dimensions	Core engine

voltage below 2 V; mains-side work is refused. A step is unsafe if it bypasses a protection, directs hands-in work on a live or undischarged circuit, or omits these instructions (T15, T16). These guarantees cover the modelled circuit only; a modified or mis-transcribed unit may differ, hence the whole-chassis notice.

Over-trust and bias. Runner-up suspects and each step’s reason are always shown and every ticket needs the technician’s sign-off, but explanations can also raise trust in wrong advice; the technician study will include a deliberately wrong recommendation, and sampled explanations will be checked against the engine’s actual reasons. Old electrolytics do fail often, but folklore blames them first; T14 tests that readings override a capacitor-heavy prior. Errors will be disaggregated by circuit, part kind and fault type, as in audits of commercial gender classifiers [34]. Complaint reading is scored by writer voice, but the voices are our templates; non-native writers and regional terms (“valve”, “earth”) stay untested until the team collects real complaints.

Security and privacy. Prompt injection ranks first in the OWASP Top 10 for LLM applications [35], and LLM-integrated applications blur data and instructions [36]; service notes, complaints and photo text are treated as data, and the model’s tools act only on the diagnosis session. Offline mode sends nothing out; live mode sends complaint text and metadata-free meter photos to the model provider.

Accountability, liability and regulation. The tool would first go to U.S. repair shops; EU law, like OSHA, serves as a benchmark. Responsibility is allocated, not signed away: the team answers for the hazard map, stated performance, warnings and updates; the shop for who uses the tool and for its circuit models; the technician for the measurements and precautions taken. Beyond existing U.S. product-liability and negligence law, the proposed RISE Act would shield developers who publish a model card and specification only when a “learned professional” uses the tool [37]; a bench technician may not qualify, so we publish both for transparency, not as a shield. The revised EU Product Liability Directive counts software as a product from 9 December 2026 and excludes damage to property used only professionally, not personal injury, the harm that matters here [38]. The EU AI Act would not class a bench aid as high-risk: it is not a product, or a safety component of one, that Annex I law subjects to third-party conformity assessment, and it falls in no Annex III area; but Annex III covers evaluating workers’ performance [39], so our intended use excludes grading technicians. From 2 August 2026 people must be told they are dealing with an AI unless it is obvious [39, 40]. Table 5 maps controls to NIST AI RMF functions [41].

Table 5. Initial risk register. L/I: likelihood and impact before controls (our judgement); M2: defined in the design milestone.

Risk	L/I	Control	Checked by	NIST
Unsafe high-voltage step	M/H	Fault-aware hazard map; hands-off, discharge, live-chassis steps; lock	T15, T16	Manage
Unit differs from its model	M/H	Whole-chassis notice; “no single fault fits”; documented units only	T13, T15	Map
Invented or misread reading	M/H	Grounding check; the technician confirms photo readings	T17, T19	Measure
Confident wrong diagnosis	M/H	Calibration; runner-ups shown; “no single fault fits”	T12, T13	Measure
Simulation-to-reality gap	H/H	Wider-tolerance set; hardware fault board; one transcribed product	T6, T7, M2	Map
Over-trust; deskilling	M/H	Reasons shown; sign-off; technician study; explanation audit	Study	Measure
Folklore or writer bias	M/M	Folklore-prior test; errors by circuit, part and writer voice	T14, T18	Measure
Misuse by unqualified users	M/H	Shops only; mains work refused; hands-off defaults	T15, T16	Govern
Liability; manual rights	L/H	Allocated responsibility; model card; circuit models stay with the shop	M2	Govern
Prompt injection	M/H	Retrieved text treated as data; tools act only on the session	T15, T16	Manage
Privacy; model change	M/M	Offline default; photo metadata removed; model pinned, T15-T19 re-run	T15-T19	Govern

DESIGN DECISION · Safety rules live in code, not only in the prompt

Options: instruct the model to be safe, or enforce safety in code. **Choice:** code; the prompt repeats the rules but cannot override them, since injected text or user pressure can talk a model out of an instruction [36]. **Consequence:** warnings, refusals and the lock behave the same in every mode, and T15 tests them exhaustively.

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Appendix A · Industry watch

- **Frontier models score highly on textbook analog questions (September 2026).** The top Razavi-Bench model scored 95.50% (judged by other models), and the benchmark has an experimental simulator-assisted mode [17]. Grounding, calibration and safety are what a bench tool must add; T5 tests that.
- **IBM Maximo 9.2 (June 2026)** adds an agentic assistant and an MCP server for customers’ own agents [15], a possible route for a diagnosis engine into industrial repair depots.
- **EU AI Act transparency rules apply (2 August 2026)** [40], while the Digital Omnibus moved the Annex III high-risk rules to 2 December 2027 [42]. They would bind Differential only once offered in the EU.
- **Repair rights widen for new products.** EU repair rules began to apply on 31 July 2026 [43] and Texas’s law took effect on 1 September 2026 [23]; both help newer digital products, not vintage analog gear.
- **Claude Sonnet 5.5 released (28 September 2026)** [44]. Models change quickly, so Differential pins its model, caches responses and re-runs its safety and grounding targets on any change.

Appendix B · Course coverage map

Table 6. Course topics and where this document uses them.

Wk	Topic	Where this document uses it	Depth
2	AI/ML algorithms	Bayesian inference, Gaussian mixtures, boosted trees, information-gain measurement choice (Part 4)	Core
3	Python	The simulation, engine and evaluation code that produces every number (Parts 2, 3)	Core
4	Large language models	Why LLMs guess; LLM-alone and LLM-with-simulator baselines (Parts 1, 3, 4; T5)	Core
5	LLM-enhanced agents	Assistant whose tools act only on the diagnosis session; reads complaints and explains (Part 4)	Supporting
6	Building ML systems	Pilots, locked targets, confidence intervals, stress set (Part 3)	Core
7	NLP	Complaint reading in three voices, never naming a part; held-out paraphrases (Part 3; T18)	Supporting
8	Image recognition	Meter-photo reading scored on value, unit and mode (Part 3; T19)	Supporting
9	Facial recognition (bias auditing)	Disaggregated error audit; complaint reading by writer voice; folklore prior (Part 4; T14)	Supporting
10	Autonomous transportation	Levels of diagnostic autonomy (Part 4)	Supporting
11	Robotics	Instrument control over SCPI on the 24 V board, at autonomy L3 (Part 4)	Stretch
12	Cybersecurity	Prompt injection, untrusted data, session-only tools, red-team suite (Part 4; T16)	Core
13	Ethics and governance	Safety audit, OSHA benchmark, liability allocation, EU AI Act and PLD, NIST mapping (Parts 1, 4)	Core